From Ghost Echoes to Useful Geometry: Covariance-Aware Virtual-Aperture Radar Tracking

***Abstract*—Specular radar echoes can provide additional observation directions, but their benefit depends on coordinate consistency, uncertain reflectors and correct measurement reuse. We derive the equivalence of sensor-image and target-image rays, explain a 15.189-degree receiving-frame mismatch, and propagate reflector distance and orientation uncertainty into coupled range/range-rate covariance. A fixed-prior iterated update relinearizes without repeatedly consuming the same evidence. In an 80-seed independent-truth range/rate tracking study, mean trajectory RMSE decreases from 2.858 m for direct observations to 0.304 m with a virtual leg and approximately 0.284 m with reflector uncertainty propagation. Diagonal and coupled propagation are nearly indistinguishable in this fixture. An earlier complete-chain prototype and 155 instrumented real-data replay records are reported separately, with metric and modeling limitations made explicit.**

***Index Terms*—automotive radar, multipath, virtual aperture, data association, covariance propagation**

I. Introduction

A guardrail echo is inconvenient precisely because it is structured. It often follows the same target through several scans, carries a stable path geometry and changes Doppler with the target's motion. Deleting it removes a false object from a conventional tracker, but can also discard a second direction from which the real object has been observed. The challenge is to recover that information without turning every coincidental echo into evidence. We organize the problem around a simple chain: estimate the reflecting geometry, test candidate paths against kinematics, and decide how much information each admitted observation should contribute. The last step matters as much as the first. A duplicated or overconfident reflection can make the filter less accurate even when its geometric interpretation looks plausible. This paper develops a receiving-frame-consistent image model, its observability and reflector-covariance consequences, and a fixed-prior iterated update that consumes each observation once. A matched 80-seed simulation uses independent truth geometry and compares direct-only tracking with three virtual-aperture covariance policies. We also reproduce a historical estimate–gate–assimilate experiment, but identify its metric and model limitations rather than promote its 67% figure into a fleet-accuracy claim. Instrumented replay counters from 155 real-data records provide a separate integration check.

II. Related Work

Multipath exploitation is not a new premise. Virtual-anchor positioning and tracking already use reflected paths as information, with uncertainty modeling central to reliable inference [6]. In automotive MIMO radar, Zheng *et al.* exploit the difference between departure and arrival angles to detect multipath ghosts using a GLRT and continuous-domain angular estimation [5]. That signal-level detection problem is distinct from updating a target state with an already hypothesized reflected observation. Kalman filtering [1] supplies the estimation machinery, while IMM tracking [2] addresses maneuver-model uncertainty. Adaptive CFAR [4] addresses clutter-dependent detection. Our contribution is not to replace these foundations or claim that image-source geometry was previously incorrect. It is to make the coordinate map, reflector uncertainty and measurement lifecycle explicit in an automotive tracking formulation and evaluate their consequences under controlled comparisons.

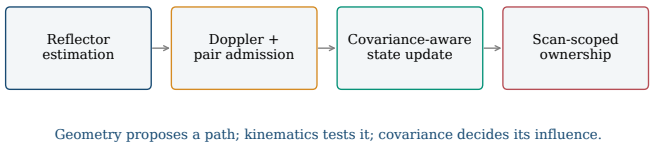


Fig. 1. The proposed processing chain separates path discovery, physical admission, statistical weighting and observation ownership. An echo is not useful merely because it resembles a reflection.

III. Geometry: Two Correct Images, One Receiving Frame

Let a planar reflector satisfy $\mathbf{n}^T\mathbf{x} = d$ with unit normal $\mathbf{n}$. Define $H = I - 2\mathbf{n}\mathbf{n}^T$ and

$$F(\mathbf{x}) = H\mathbf{x} + 2d\mathbf{n}. \tag{1}$$

For sensor $\mathbf{s}$ and target $\mathbf{p}$, their images are $\mathbf{s}^* = F(\mathbf{s})$ and $\mathbf{p}^* = F(\mathbf{p})$. The two unfolded rays obey

$$\mathbf{p}^* - \mathbf{s} = H(\mathbf{p} - \mathbf{s}^*). \tag{2}$$

Because $H^T H = I$, their lengths agree exactly. Their coordinates are not interchangeable as physical arrival bearings: the Householder map must be applied when returning an unfolded ray to the receiving frame.

A. The 15.19-degree example, stated precisely

In Fig. 2, the unfolded sensor-image direction is $(30, -4)$ and the target-image direction is $(30, 4)$. Confusing their coordinates gives

$$\Delta\theta = 2\arctan(4/30) = 15.1893^\circ. \tag{3}$$

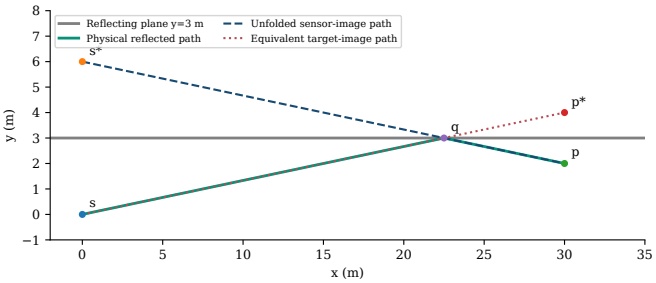


Fig. 2. Planar example with sensor $(0, 0)$ m, wall $y = 3$ m and target $(30, 2)$ m. The physical contact is $(22.5, 3)$ m. Sensor-image and target-image constructions have the same length and are related by a reflection; using one ray's coordinates as the other's bearing creates a sign error.

At the longitudinal plane $x = 30$ m, the corresponding ray endpoints differ by 8 m laterally. This is a frame-mismatch example, not a universal defect in target mirroring and not a constant error at arbitrary 30 m slant range. The propagation class must also be specified. A reflected one-way leg has length $\rho_s = \|\mathbf{p} - \mathbf{s}^*\|$. A mixed direct/reflected radar round trip has total length $\rho_d + \rho_s$ and conventional apparent range $(\rho_d + \rho_s)/2$. A symmetric two-reflected-leg path has apparent range ρ_s. The numerical study below evaluates a calibrated virtual-leg observation model; applying it to an unclassified mixed path would introduce model error.

B. What the displacement invariant does and does not prove

For an origin-centred sensor, define $\mathbf{q}_s = \mathbf{p} - 2d\mathbf{n}$. Then

$$\mathbf{p} - \mathbf{q}_s = 2d\mathbf{n}. \tag{4}$$

Several independent estimates of $\mathbf{q}_s$ for a common plane share a rank-one displacement direction. If $\mathbf{q}_s$ is constructed from the same assumed $(d, \mathbf{n})$, however, the equality is an identity rather than an association test. A useful residual must compare independent observations after a physically justified coordinate conversion. Ordinary target-image differences instead have magnitude $2|\mathbf{n}^T\mathbf{p} - d|$, which varies with target location.

IV. A Virtual Aperture and Its Information

For state $\mathbf{x} = [\mathbf{p}^T, \mathbf{v}^T]^T$, origin-centred sensor and static reflector, define

$$\rho_d = \|\mathbf{p}\|, \qquad \dot{\rho}_d = \mathbf{u}_d^T\mathbf{v}, \tag{5}$$

$$\rho_s = \|\mathbf{p} - 2d\mathbf{n}\|, \qquad \dot{\rho}_s = \mathbf{u}_s^T\mathbf{v}, \tag{6}$$

where $\mathbf{u}_d = \mathbf{p}/\rho_d$ and $\mathbf{u}_s = (\mathbf{p} - 2d\mathbf{n})/\rho_s$. A parent and its reflection generally do not share a Doppler bin: $\Delta\dot{\rho} = \mathbf{v}^T(\mathbf{u}_s - \mathbf{u}_d)$. The project fixture produces a 1.987 m/s difference for a lateral target velocity of 10 m/s; that is large compared with its 0.05 m/s radial-speed noise.

A. When does the second direction help?

The range-position Jacobian is $U = [\mathbf{u}_d^T; \mathbf{u}_s^T]$. In two dimensions,

$$\det(U^TU) = \sin^2\psi, \tag{7}$$

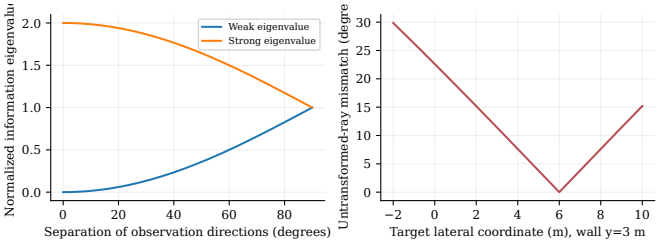


Fig. 3. Analytical geometry diagnostics. Left: information eigenvalues expose the weak direction as the two paths become parallel. Right: untransformed bearing mismatch varies with target position; it is not a universal constant.

where ψ is the angle between the directions. With equal independent range noise σ^2, the normalized information eigenvalues are $1 \pm |\cos\psi|$. Two nearly parallel paths add mostly radial information; they do not magically supply strong lateral accuracy. The second direction becomes useful when its angular separation is appreciable and its noise model remains credible.

V. Reflector Uncertainty and a Single-Use Update

Parameterize a planar normal as $\mathbf{n} = (\cos\phi, \sin\phi)^T$ with tangent $\mathbf{t} = (-\sin\phi, \cos\phi)^T$. For $\boldsymbol{\xi} = (d, \phi)^T$ and independent reflector uncertainty Σ_ξ, linearization gives

$$R_{\text{eff}} = R_0 + J_\xi\Sigma_\xi J_\xi^T. \tag{8}$$

The specular-range derivatives are $\partial\rho_s/\partial d = -2\mathbf{u}_s^T\mathbf{n}$ and $\partial\rho_s/\partial\phi = -2d\mathbf{u}_s^T\mathbf{t}$. Let $\mathbf{w} = (\mathbf{v} - \dot{\rho}_s\mathbf{u}_s)/\rho_s$; the corresponding rate derivatives are $-2\mathbf{w}^T\mathbf{n}$ and $-2d\mathbf{w}^T\mathbf{t}$. Range and range rate therefore share reflector-induced covariance. Inflating their diagonal entries independently discards that correlation.

A. Iterate the linearization, not the evidence

Let $(\mathbf{x}^-, P^-)$ be the predicted prior. At iteration j, compute $h_j = h(\mathbf{x}_j)$, H_j and R_j, then

$$K_j = P^-H_j^T(H_jP^-H_j^T + R_j)^{-1}, \tag{9}$$

$$\mathbf{x}_{j+1} = \mathbf{x}^- + K_j[\mathbf{z} - h_j + H_j(\mathbf{x}_j - \mathbf{x}^-)]. \tag{10}$$

The prior covariance remains fixed during these iterations. After the final linearization, update covariance once using the Joseph form. Repeatedly applying an ordinary measurement update and shrinking covariance inside the same iteration loop treats the same data as fresh independent evidence. The distinction is consequential when range-rate linearization couples a poor velocity prior into a position innovation.

B. Ownership and suppression

Every admitted observation receives an identifier scoped to sensor, scan and detection. Assimilation marks that identifier consumed. Downstream multipath suppression must not penalize the same evidence after the target update used it. The implementation's negative sentinel is one representation of this contract; an explicit enum is another. The research requirement is the lifecycle, not the numerical value of a private field. A reused detection must be excluded or modeled with its correct correlation.

VI. From a Plane to a Curved Barrier

Let $\mathbf{q} = \mathbf{s} + t\mathbf{u}$ be a specular contact, $\boldsymbol{\delta} = \mathbf{p} - \mathbf{s}$, and $\ell = t + \|\boldsymbol{\delta} - t\mathbf{u}\|$ the broken-path length. Rearrangement yields

$$t = \frac{\ell^2 - \|\boldsymbol{\delta}\|^2}{2(\ell - \mathbf{u}^T\boldsymbol{\delta})}. \tag{11}$$

Require a unit $\mathbf{u}$, positive denominator and $0 < t < \ell$. Grazing and collinear configurations are poorly conditioned even when the formula returns a finite number. Two contacts with tangent slopes determine a local cubic Hermite graph. Its confluent Vandermonde determinant is $(x_2 - x_1)^4$, establishing uniqueness for distinct abscissae but not numerical stability. A shifted, scaled coordinate is preferable. This graph is a local approximation, not an exact clothoid.

VII. Association Under Maneuvers

For a candidate residual $\mathbf{y}$ with innovation covariance S, the whitened statistic is $D^2 = \mathbf{y}^T S^{-1}\mathbf{y}$. A bounded kinematic policy may use

$$\gamma = \mathrm{clip}_{[\gamma_0, \gamma_{\max}]}\{\gamma_0(1 + c_a a_n + c_e D^2/m)\}, \tag{12}$$

where a_n is normalized acceleration and m is measurement dimension. Unlike a ratio of already-whitened D^2 to a dimensional covariance trace, each added term is dimensionless. The policy depends on the tested candidate, so a fixed chi-square coverage statement no longer applies automatically. For $c = \gamma_0 c_e/m < 1$, its unsaturated acceptance inequality becomes $D^2 \le \gamma_0(1 + c_a a_n)/(1 - c)$; if $c \ge 1$, it admits every candidate until the outer clamp acts. This identifies an "open-door" failure analytically. An IMM can predict mode probabilities before association [2]. A narrow common gate may nevertheless starve a maneuver mode of evidence. That is a useful onset stress test, not a universal refutation of IMM tracking. The archived adaptive-gate prototype also propagates the tracker once per attempted candidate, which confounds candidate order with elapsed time; its 99.6% result is consequently retained as exploratory rather than used as the headline comparison here.

VIII. Experimental Evaluation

A. Independent-truth covariance study

We simulate a two-dimensional constant-velocity target from $\mathbf{x}_0 = (25, 1, 4, 0.4)^T$, a true wall at $d = 3$ m and $\phi = \pi/2$, 60 scans at 20 Hz, range noise 0.10 m and radial-speed noise 0.05 m/s. The filter starts with component perturbation standard deviations $(1, 1, 2, 2)$ and matching diagonal covariance. Discrete process noise is $GG^T(0.3)^2$, with $G = [\frac{1}{2}\Delta t^2 I; \Delta t I]$. Each scan supplies an external reflector estimate with distance standard deviation 0.15 m and angle standard deviation 2°. Measurements are generated from the true reflector, never from the estimate supplied to the filter. Four methods share each seed's initialization and measurement draws: direct only, dual with sensor-noise covariance only, dual with diagonal reflector propagation, and dual with full coupled propagation. Every filter uses three fixed-prior iterations and one covariance update. For each seed we calculate $\sqrt{K^{-1}\sum_k \|\hat{\mathbf{p}}_k - \mathbf{p}_k\|^2}$, then average this RMSE across 80 seeds. Confidence intervals bootstrap complete seeds, not correlated individual scans. The mean RMSE is 2.858 m for direct-only tracking, 0.304 m for dual observations with sensor noise only, 0.284 m for diagonal uncertainty propagation and 0.284 m for coupled propagation. The result supports the value of the additional path in this deliberately weak direct-only range/rate geometry. It does not imply the same gain over a tracker already using accurate angle measurements. Nor does it establish superiority over a modern automotive tracking baseline.

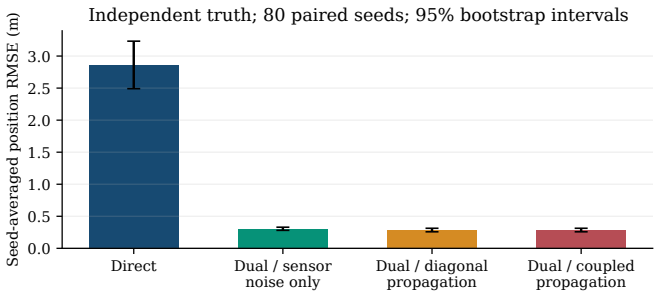


Fig. 4. Matched-seed independent-truth comparison. Error bars are 95% percentile bootstrap intervals over complete seeds. The diagonal and coupled policies are nearly indistinguishable in this fixture; the data do not support a large full-covariance gain over diagonal propagation.

TABLE I
Independent-truth position RMSE across 80 seeds

Method	Mean (m)	95% interval (m)
Direct only	2.858	[2.491, 3.232]
Dual / sensor noise	0.304	[0.281, 0.330]
Dual / diagonal	0.284	[0.258, 0.313]
Dual / coupled	0.284	[0.258, 0.313]

B. Reproducing the earlier complete-chain result

The existing 40-seed experiment recovers edge recall/precision 0.969/0.902 and mean Euclidean position error 1.969 to 0.651 m. The distance/normal errors average 0.260 m and 3.282°. Three details limit interpretation. First, the reported position statistic is mean Euclidean error, not RMSE. Second, the tracking leg generates ghost observations using the supplied estimated reflector, so it does not expose the complete geometric mismatch to the filter. Third, it repeats covariance updates inside a three-step loop. We report this result to connect the prototype history to the present controlled study, not as independent validation of a robust end-to-end fleet system.

C. Instrumented real-data replay

The 155 paired replay records contain 71,779 detections, 9,478 OFF versus 9,131 ON association events, and 10,130 ON assimilation marks in 151 records. These are reproducible counters, not labeled association accuracy. The harness constructs pairing candidates from geometric proximity rather than

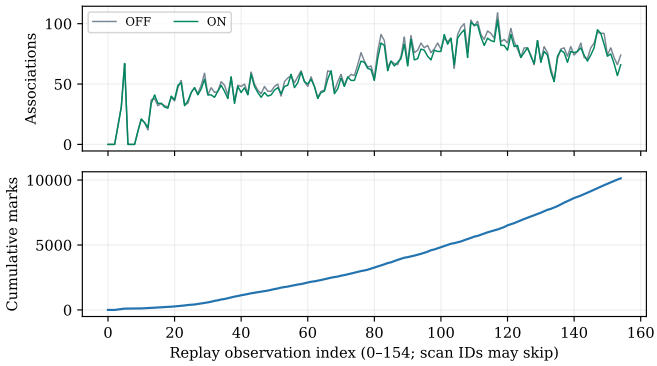


Fig. 5. Actual instrumented replay: association counts and cumulative assimilation marks. Scan IDs are not consecutive; the horizontal coordinate is the paired record index. This figure tests event accounting, not ground-truth tracking accuracy.

independently annotated reflection paths. Neither position truth nor recall denominators are present in those CSVs.

IX. Discussion and Conclusion

Multipath is valuable when it supplies an additional, sufficiently different observation direction and when uncertainty follows the geometry into the update. A correct image construction alone is not enough. Receiving-frame conversion, independent truth generation, single-use evidence and a clear distinction between detection counters and target accuracy are equally necessary. The strongest result here is a transparent controlled study: virtual-leg observations reduce error in a range/rate-only tracking geometry, while reflector uncertainty modestly improves their weighting. Full covariance is mathematically justified, but its incremental benefit over diagonal propagation is small in the tested fixture. Curved surfaces, mixed propagation classes, uncertain association and comparison with an angle-enabled production tracker remain the meaningful next tests. That is a sharper foundation for a conference paper than attributing every improvement in a large software stack to a single impressive percentage.

References

[1] R. E. Kalman, “A new approach to linear filtering and prediction problems,” *Journal of Basic Engineering*, vol. 82, pp. 35–45, 1960, doi:10.1115/1.3662552.

[2] H. A. P. Blom and Y. Bar-Shalom, “The interacting multiple model algorithm for systems with Markovian switching coefficients,” *IEEE Trans. Automatic Control*, vol. 33, pp. 780–783, 1988, doi:10.1109/9.1299.

[3] S. M. Kay, “A fast and accurate single frequency estimator,” *IEEE Trans. Acoustics, Speech, and Signal Processing*, vol. 37, pp. 1987–1990, 1989, doi:10.1109/29.45547.

[4] H. Rohling, “Radar CFAR thresholding in clutter and multiple target situations,” *IEEE Trans. Aerospace and Electronic Systems*, vol. AES-19, pp. 608–621, 1983, doi:10.1109/TAES.1983.309350.

[5] L. Zheng, J. Long, M. Lops, F. Liu, and X. Hu, “Detection of ghost targets for automotive radar in the presence of multipath,” *IEEE Trans. Signal Processing*, 2024, doi:10.1109/TSP.2024.3384750.

[6] E. Leitinger, P. Meissner, and K. Witrisal, “Cognitive indoor positioning and tracking using multipath channel information,” arXiv:1610.05882, revised 2021.

[7] S. Särkkä and Á. F. García-Fernández, “Temporal parallelization of Bayesian smoothers,” arXiv:1905.13002, 2020.